End of life challenges associatted with Polyoelefins (PO)

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Outline

1 introduction

2 reactivity of Alkene and Polyalkene

3 Challenges with End of Life

4 Conclusion and Outlook

Definition of Polyolefins (PO)

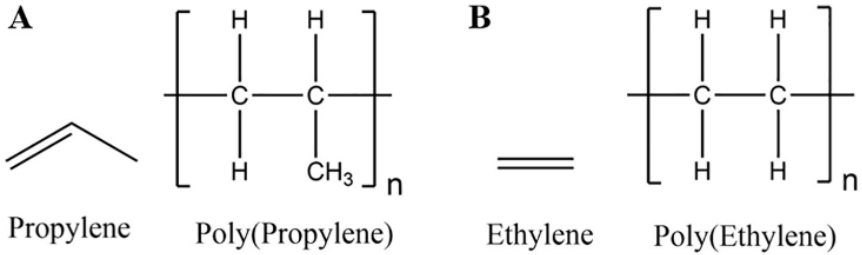


Figure 1. *Monomer and polymer of (A) PP and (B) PE [1].*

PO represent 40% of the world plastic production

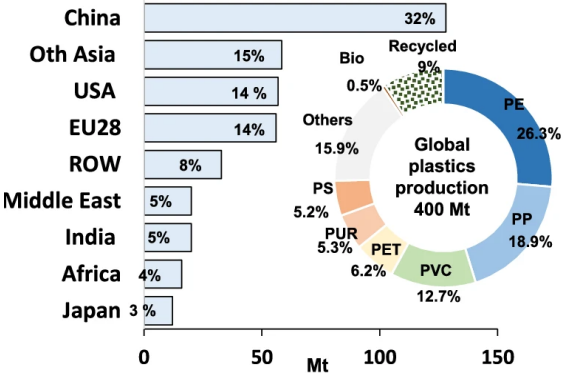


Figure 2. *Global plastic production breakdown by region/type [2]*

Why polyolefin are so popular ?

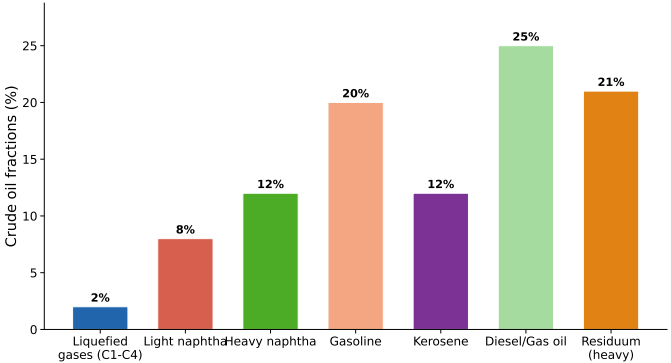


Figure 3. *Atmospheric distillation of crude oil [3]*

Naphtha → Ethylene, propylene **or** gasoline (oct. 95 or+) depending on the **Market**.

because the raw materials are cheap

Naphtha + HO (steam) → Furnace (850°C) → Ethylene (30%) + Propylene (16%) + H_2 + CH_4 + C4s + BTX + coke

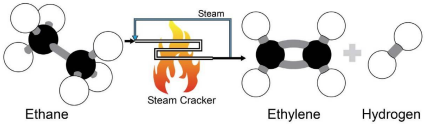


Figure 4. *Formation of ethylene from ethane (C_2H_6) [4]*

natural gas (95% CH_4 + 3% C_2H_6)

can also be converted into Ethylene with higher yield (shale gas of US Gulf cost). [5]

Because PO are versatile

PP (Polypropylene)

- M_w
- Tacticity

PE (Polyethylene)

- M_w
- PDI
- Branching

for thermoplastics

$M_w +$ PDI $+$ branching $+$ Tacticity $= \xi_c$ & properties (E,ρ σ and T_g)

trade off

PE density knob $\rightarrow$ HDPE pipes vs LDPE films
PP tacticity knob $\rightarrow$ Homo (stiff) vs Copo (tough)

PO exhibit excellent chemical resistance

Peu important	Alcanes et composés halogénés
	Éthers et thioéthers
↓	Alcènes et alcynes
	Amines
	Alcools et thiols
	Cétones
	Aldéhydes
	Nitriles
	Amides
	Esters
	Halogénures d'acide
	Acides carboxyliques
Très important	Cations

Figure 5. *Functional group reactivity (adapted from LMAPR1230).*

Summary of PP/PE properties

PE (Polyethylene) [6]

Key Advantages

- Low density: 0.91–0.97 g/cm^3
- Chemical resistance: Excellent (acids, bases, solvents)
- Temp range: -100°C to +130°C (HDPE), +80°C continuous
- Processable: Injection, blow molding, extrusion, film blowing
- Cost: $1,200–1,800/tonne (2026 avg)
- **Tunable**: Density (stiffness), MFI 0.1–50

PP (Polypropylene) [6]

Key Advantages

- Low density: 0.90–0.91 g/cm^3
- Chemical resistance: Excellent (acids, bases, oils)
- Temp range: 0°C to +140°C
- Processable: Injection molding, extrusion, fibers
- Cost: $1,100–1,600/tonne (2026 avg)
- **Tunable**: MFI 0.3–100, copolymer % (impact)

PO is mainly used for short-term usage

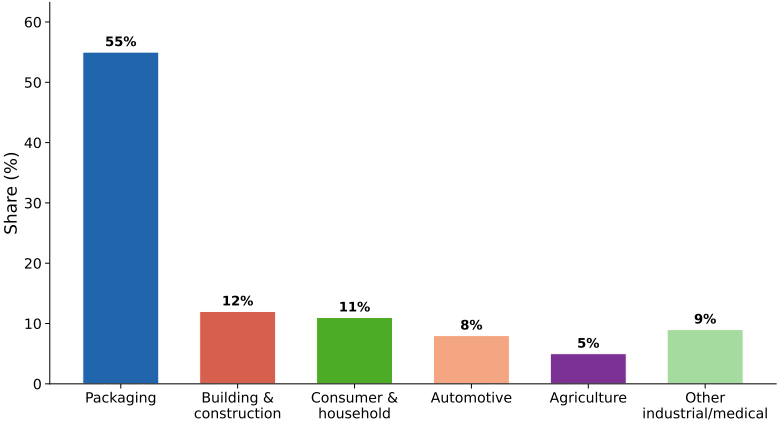


Figure 6. *PO per final use [7]*

Alkene's reactivity due to the π bond

Most reactions of alkenes involve additions to this π bond, forming new single bonds.

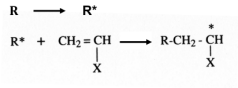
- addition with a nucleophile
- **addition with a radical**
- Oxidation reaction

Alkenes can polymerize by Chain-reaction Polymerization [8]

- **Free Radical Polymerization (FRP)**
- **Complex coordination polymerization**

FRP : main reaction scheme

Initiation



R* = radical, anion or cation

Propagation

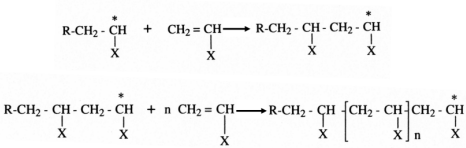


Figure 7. *FRP reaction scheme.*

FRP: Parameters Influencing Stereochemistry

Most vinyl polymers from free radical polymerization (FRP) exhibit **atacticity** due to the planar sp^2 radical center allowing random addition from either face [8].
Key parameters influencing stereoselectivity include:

- **Temperature**: Lower temperatures favor syndiotacticity via restricted rotation and steric control.
- **Steric hindrance**: Bulky R-groups increase syndiotacticity through steric hindrance.

PE & PP

PE is not concerned. **PP** is strictly atactic in FRP

Radical transfer between two chains lead to long branches

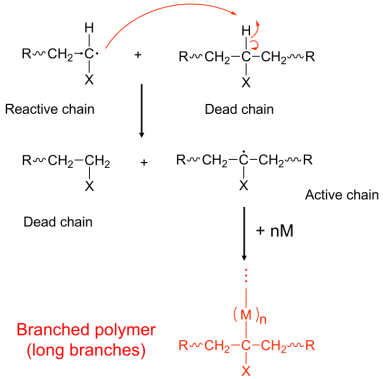


Figure 8. *Inter-chain transfer [8].*

→ long branches in PE.

Radical transfer within one chain lead to short branches

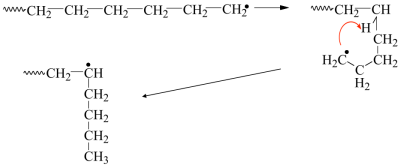


Figure 9. *Intra-chain transfer [8].*

→ short branches in PE.

Transfer to monomer

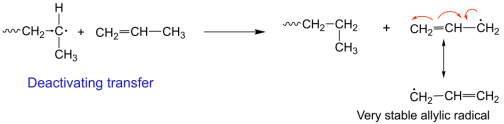


Figure 10. *Chain neutralization in PP [8].*

→ High molecular weight PP is not achievable in standard conditions.

FRP : conclusion

Only LDPE

- High T and high P
- Amorphous ← a lot of branches of different sizes
- PDI = 7-20
- Application: Soft plastic bag

Only atactic PP

- Application: Adhesive (almost never produced)

Complex coordination polymerization: a revolution

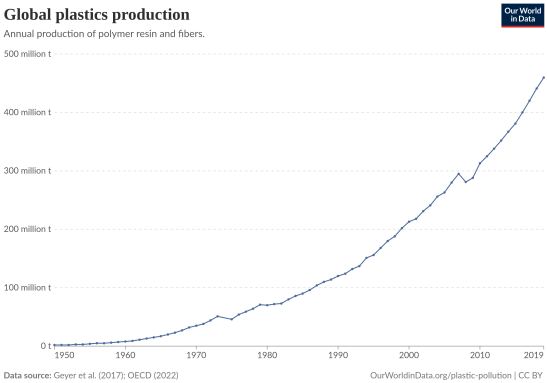


Figure 11. *Global plastic production since 1950 [9].*

Ziegler-Natta catalyst

Catalyst:

- Heterogeneous transition metal (Ti)
- Organometallic of a metal from (I to III) (Al)

Typically:

$$TiCl_4 \text{ or } TiCl_3 + AlR_3 \tag{1}$$

with $R = CH - CH_3$ for polyethylene.

Major results with the introduction of ZN catalyst

According to De Moustier [8]

- Stereospecific polymerization of PE
- Synthesis of linear PE
- Copolymerization of PP/PE

→ Control of stereoregularity/stereochemistry and M_w through termination reactions.
→ Tacticity depends on the crystalline form of the catalyst.

Major Polyolefins made by ZN catalyst

PE

- HDPE ($\xi_c = 0.9$ & PDI = 3-5)
- UHWPE ($M_w = \pm 3 \times 10^6$)
- LLDPE (Ethylene copolymer with 5-12% of α-olefin)

Isotactic PP

- Soluble metallocene catalyst
- 99% tacticity

End of life route

- No Biodegradability
- No chemical recycling : Polymerization always $\Delta G < 0$
- Mechanical recycling
- Mismanagement
- Thermo-degradation to form fuel [1].

What happens to the polymer structure ?

According to Ragaert *et al.* [10] :

1. Re-extrusion step
2. The use phase

Degradation mechanisms:

- **Thermo-mechanical degradation.**
- Thermo-oxidative degradation.
- Hydrolytic degradation.
- **Impurities and contaminants.**

leading to two thermo-mechanical degradation mechanisms

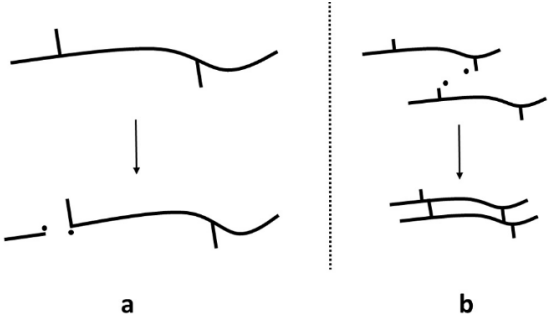


Figure 12. *Random chain scission (a) and cross-linking (b) [10].*

$M_w \downarrow$ **or** $M_w \uparrow$ and $M_w \propto E$, σ_Y, ϵ_{break}, σ_{Impact}, χ_c and μ.

Polyolefins behave very differently during mechanical recycling.

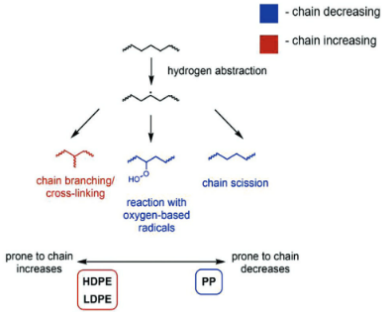
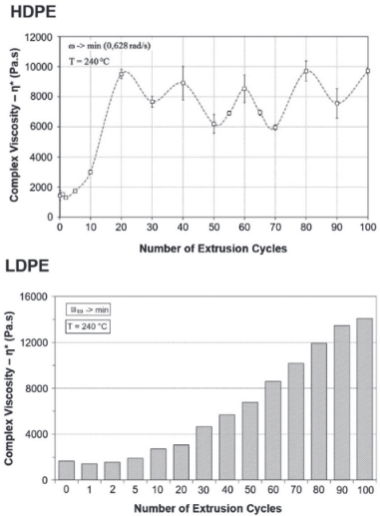


Figure 13. *General reaction scheme of polyolefins undergoing extrusion.*



Effect of Recycling on Polyolefin Properties

Property	Typical Effect	HDPE	PP
Molecular Weight	↓ (chain scission)	vary	↓ Significant
PDI	Changes	↑ Increases	↓ Decreases
Tensile Strength	↓ Loss	10-30%	10-30%
Crystallinity	↓ Loss	5-10%	5-10%
Melt Flow Index (MFI)	↑ Increases	More fluid	More fluid
ESCR	↓ Loss	60% reduction	—

Table 1. *

Loss of properties [1] ESCR: Environmental Stress Crack Resistance

Pyrolysis

making oil

Chemical recycling

need catalyst and not mature

Sorting problem

PE/PP blend

Too many grades

Conclusion

PO are popular due to

- cheap raw material
- cheap process
- ability to tune the physical properties
- very stable

Thank you for your attention

Do you have any questions ?

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